Class 4 SP250: Mods Part 2

After modulus m is picked, mods are just remainders. There are m different choices for the remainder mod m. Remainders add, subtract and multiply!

Odd and Even: m=2 There are two possible remainders modulo 2, that correspond to whether a number is even or odd. All even numbers are congruent to 0 (and each other) modulo 2 and all odd numbers are congruent to 1 (modulo 2).

How to solve HW3 #6 with mods:

Note that: $x \equiv -x \pmod 2$ for any x because their difference is 2x and $2 \mid 2x$.
Hence, modulo 2, Sergey's expression is congruent to
$1 + 2 + 3 + \ldots + 8 + 9 = 45 \equiv 1 \pmod 2$ and hence it can not be zero. QED.

Review: Mod Manipulation Theorem

Let a,b,c,d and m be integers with $m \neq 0$.

a) If $a \equiv b \pmod m$ and $c \equiv d \pmod m$ then $(a+c) \equiv (b+d) \pmod m$
Remainders ADD.

b) If $a \equiv b \pmod m$ and $c \equiv d \pmod m$ then $(a-c) \equiv (b-d) \pmod m$
Remainders SUBTRACT

c) If $a \equiv b \pmod m$ and $c \equiv d \pmod m$ then $ac \equiv bd \pmod m$
Remainders MULTIPLY.

Mods, in general, cannot be canceled by division.

Example: $3x \equiv 3y \pmod 9$. Does it imply that $x \equiv y \bmod 9$? NO.
x = 1. y = 4. 3x = 3. 3y = 12. **$3 \equiv 12 \bmod 9$?** ← YES.
Is $1 \equiv 4 \bmod 9$? NO.

Mod 10 examples:

1. Example 1: What is the last digit of 2055223 + 2959238? (Remainders ADD)
   Answer: Last digit of that sum is the same as the last digit of 3+8, which is 1.
2. Example 2: What is the last digit of 32^{100}? (Remainders MULTIPLY)
   Answer: Find the pattern. Look at the last digit of 32. Powers of 2 end on
   2, 4, 8, 6, 2, 4, 8, 6, ...
   Hence the pattern has length 4 and because $100 \equiv 0 \pmod 4$, the last digit of 22^{100} is 6.
3. Example 3: What are the last 3 digits of $256 \cdot 998^2 \cdot 997$? (Remainders MULTIPLY)
   Answer: Replace 998 with (− 2) and 997 with (− 3). The expression becomes
   $256 \cdot (-2)^2 \cdot (-3) = 256 \cdot 4 \cdot (-3) = 1024 \cdot (-3)$ <−− same as (− 72).
   Hence the product ends on (− 72) or 928.

Why do these techniques work?

Answer: Last digit of a number is its remainder when divided by 10. We have been taught in school that one can operate on remainders the way we can operate on normal numbers. Hence (almost) everything works.

Question 1: There are 9 1's written on the board. Jeffrey comes and inserts a single 2 anywhere in 10 possible spots (before the first 1, between 1st and 2nd, …). Is it possible that Jeffrey's new 10-digit number is a square?

Answer: this is impossible.

A number, consisting of 9 1's and a 2, has a remainder 2 when divided by 3.

Because the sum of the digits of said number is 11, and 11 is congruent to 2 when divided by 3, hence not a square.

Question 2: What is the last digit of 7^{2026}?

7, 9, 3, 1, 7, 9, 3, 1, 7, 9, 3, 1, ….

The pattern repeats with a cycle of 4.

$2026 \equiv 1 \ (mod\ 4)$. So 7^{2026} ends with the same digit as 7^2=9.

Question 3: Nathan wrote a 2-digit number on the board. Tappan took Nathan's number and reversed the digits. What are all possible values of the absolute value of the difference between Tappan's number and Nathan's number?

Let's say Nathan's number is $\overline{ab} = 10a + b$.

Tappan's number is then $\overline{ba} = 10b + a$.

Their difference is $|(10a + b) - (10b + a)| = |9a - 9b| = 9|a - b|$.

The answer is all multiples of 9, starting from 0 and ending on 81.

Is this a complete proof?

No, it is not – in order for this to be a complete proof, one needs to demonstrate that all multiples of 9 are actually possible. By taking Nathan's number to be 90, 91, 92, …, 99, we see that they indeed are all possible.

<u>Equations in integers often use mods</u>

Do there exist integers x and y such that 6x + 15y = 28?

No because the left-hand side of the equation is a multiple of 3 and the right-hand side is not.

<u>Mods equations - classic example</u>

Find all solutions to the equation $x^2 = 2y^2$ in integers. That is, find values of x, y to make the equation true.

x=0, y=0 is a solution.

Is x even or odd?

x is even. Let x = 2z. $(2z)*(2z) = 2y^2 \Rightarrow 4z^2 = 2y^2 \Rightarrow 2z^2=y^2$.

Same equation, different variables! Substitute y = 2t. Continue.

$$2z^2 = (2t)*(2t) \Rightarrow z^2 = 2t^2.$$

… and so on.

It seems implausible that we can cancel 2's from x and y indefinitely. How do we make it rigorous?

Let's assume that there are solutions to the equation other than x=y=0.

Pick a solution with the smallest |x|.

Since we have shown that x and y are both even, |x/2| and |y/2| are both integers and are again a solution.

If $|x| > 0 \Rightarrow |x/2| = |x| / 2 < |x|$. Hence we have constructed a smaller solution, which is a contradiction.

Definition - **Rational numbers**: numbers that can be written as a fraction x/y with x and y integers and y non-zero.

Is every number rational? (in other words, is every number a fraction?)

-> We have proven that √2 is not a rational number.

Assume that √2 can be written as a fraction: $\sqrt{2} = x/y$ with y non-zero.

Square both sides:

$$2 = x^2 / y^2$$

$$2y^2 = x^2 \text{ !!! } \leftarrow \text{look familiar?}$$

Because the only solutions to that equation are x=y=0, √2 is not rational.